



“When GPS fails and RF links jam, SUTRA geonavigates and locates survivors in real-time.”

PROJECT SUTRA.

Swarm Unified Tactical Reconnaissance Architecture — decentralized multi-UAV flight, Deep JSCC neural zero-cliff video, and 3.59cm terrain-corrected DEM survivor geolocation.

01 DECENTRALIZED GNC

50Hz OFFBOARD RATE

PX4 MicroXRCE + VIO EKF2 Odometry

02 SEMANTIC WIRELESS

-5dB JAM RESILIENCE

Deep JSCC Neural Zero-Cliff Video

03 AI VISION & GEO

3.59cm SLOPE
PRECISION

Thermal AI + 3D Terrain Raycasting

CORE ARCHITECTURE TEAM (OFFGRID)

RESEARCH-BACKED & EMPIRICALLY VALIDATED

Harika

Operations Lead

PRESENTER • OPS

Vedanth Sai Ram

AI & Vision Lead

PRESENTER • VISION

Nikhil

Tech Lead · Swarm Architect

PRESENTER • GNC/COMMS

Siva Kesava

Lead · Subsystem D

3D GIS GCS

Rohith Kumar

Lead · Subsystem F

FIELD CONOPS